

Explaining Random Forests using Smoother Weights

Kai Lüdemann, Robi Bhattacharjee, Ulrike von Luxburg

Explainable AI

- Many AI models are **black boxes**.
- Transparency needed for sensitive use cases – e.g. medical diagnoses.

Problem

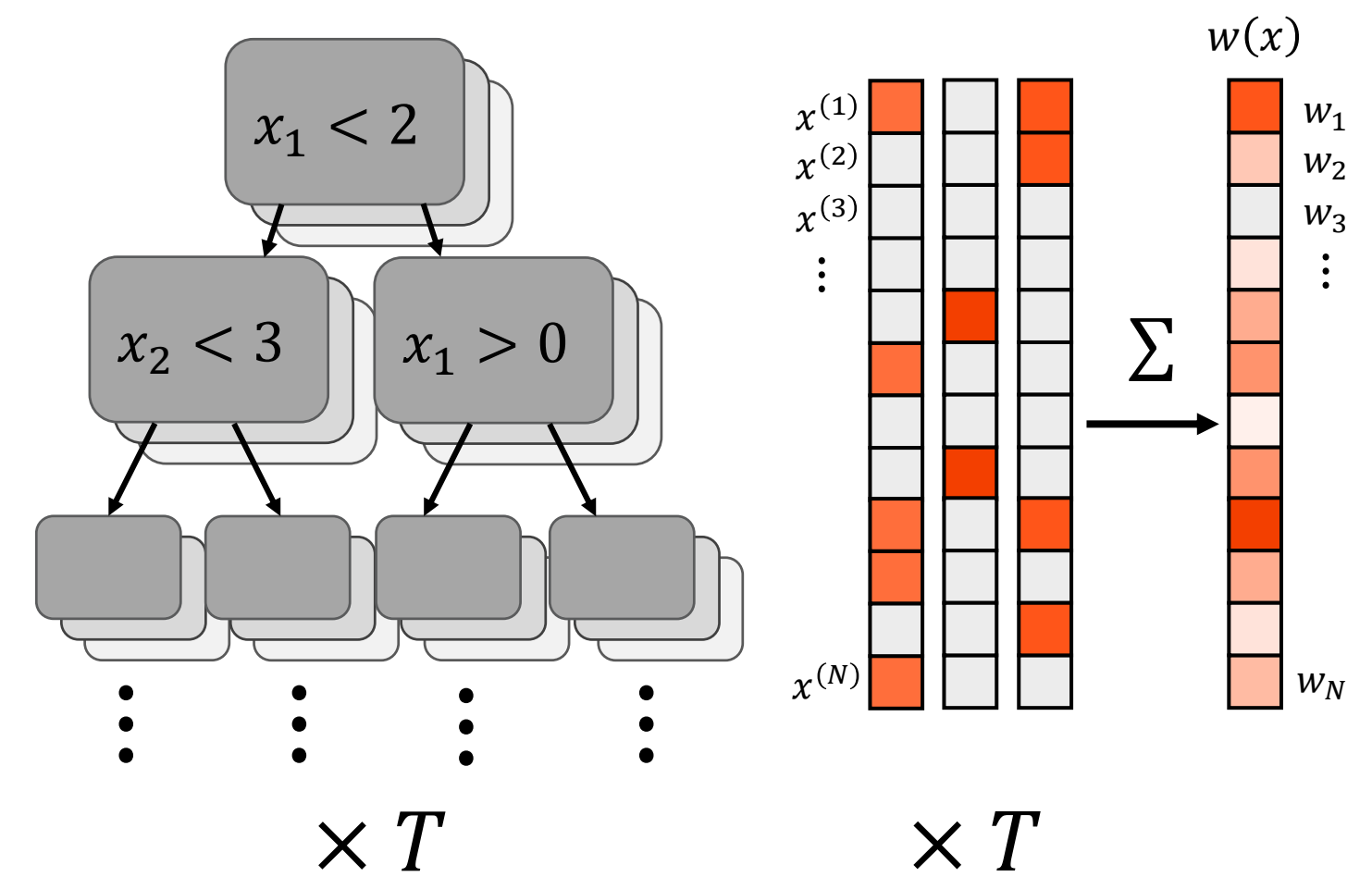
Can we produce feature importance scores for random forests from their implied smoother weights?

Random Forests as Smoother?

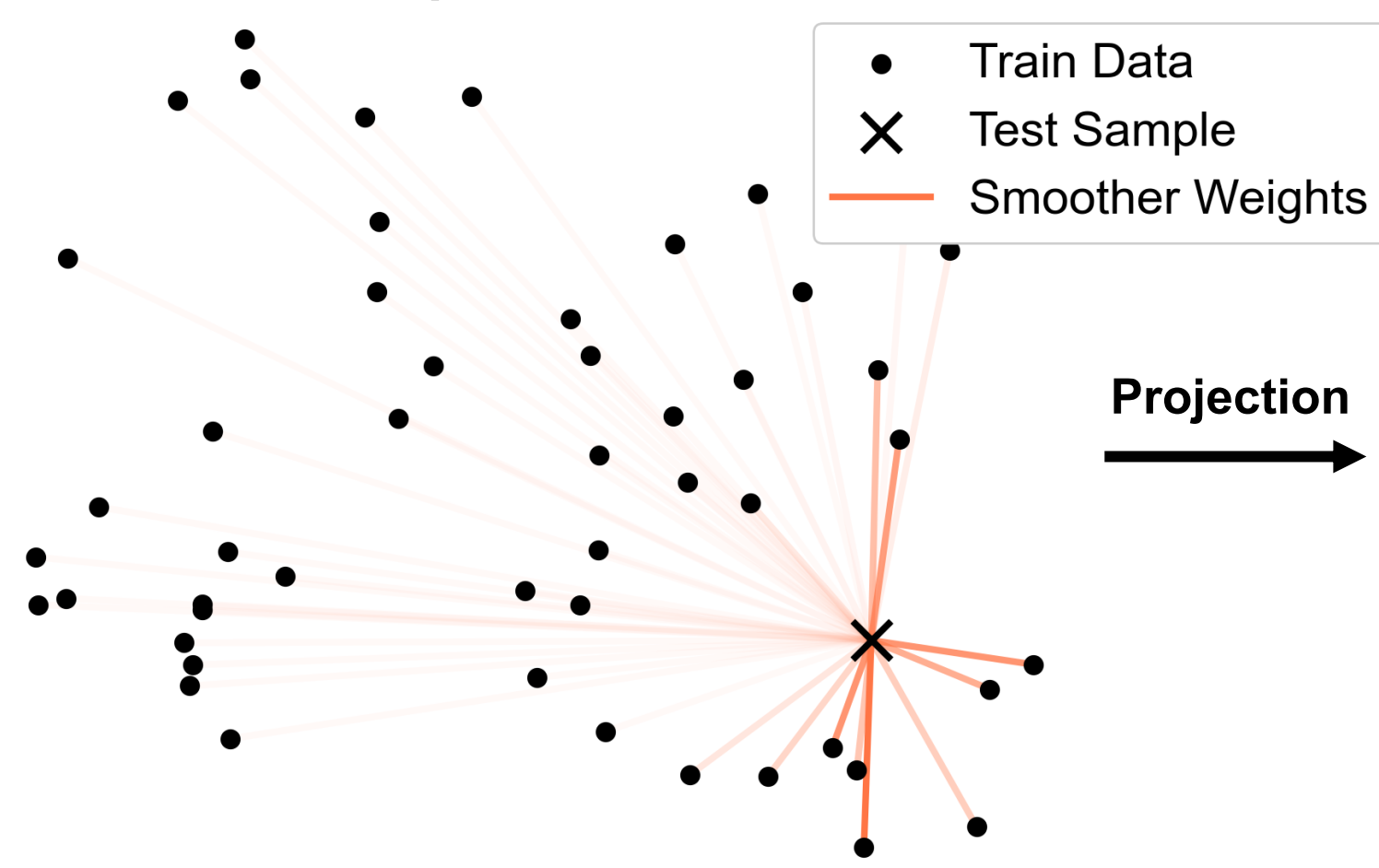
- Decision trees** average over training samples in the same leaf.
- Random forests** are ensembles of trees.
- Can compute implied smoother weights.

Smothers

$$f(x) = \sum_{i=1}^N w_i(x) y_i$$

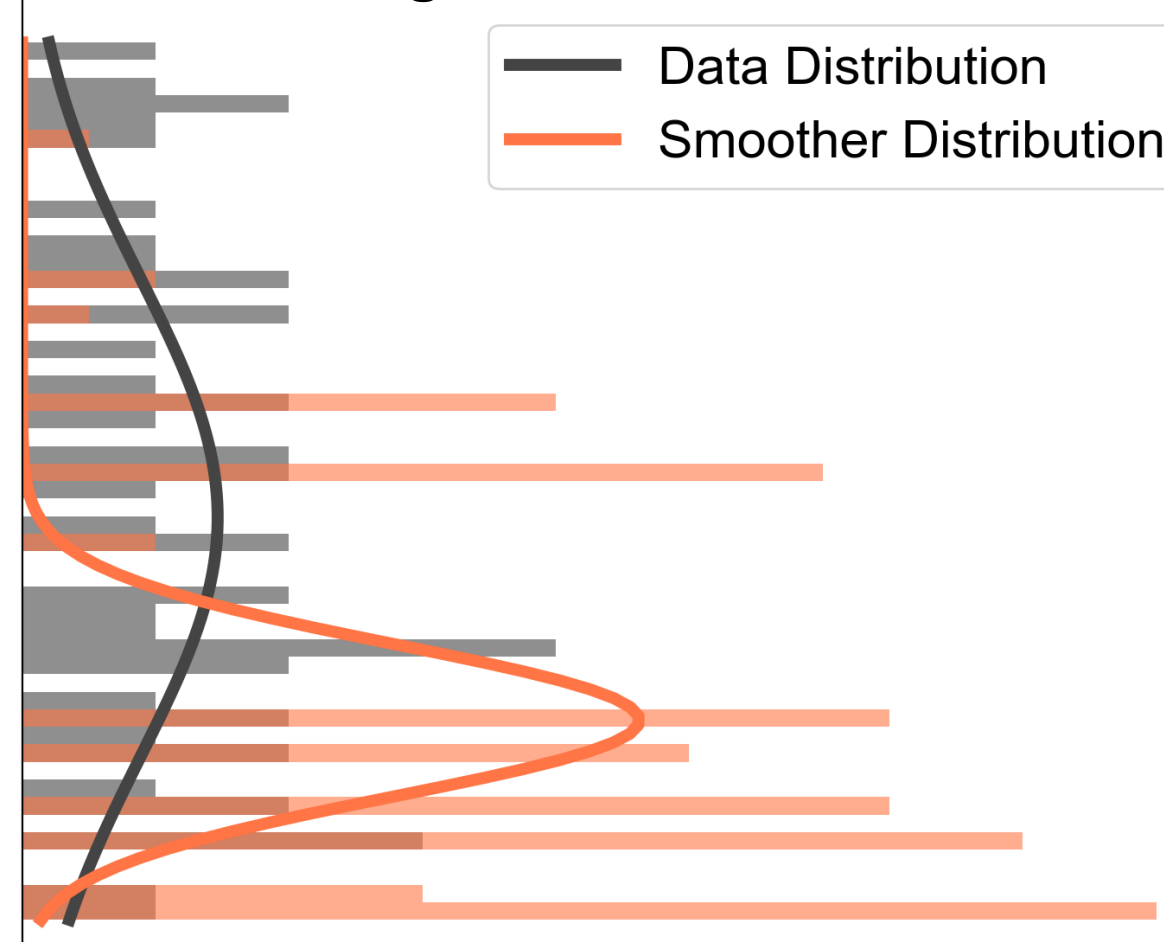


Example 2D Dataset



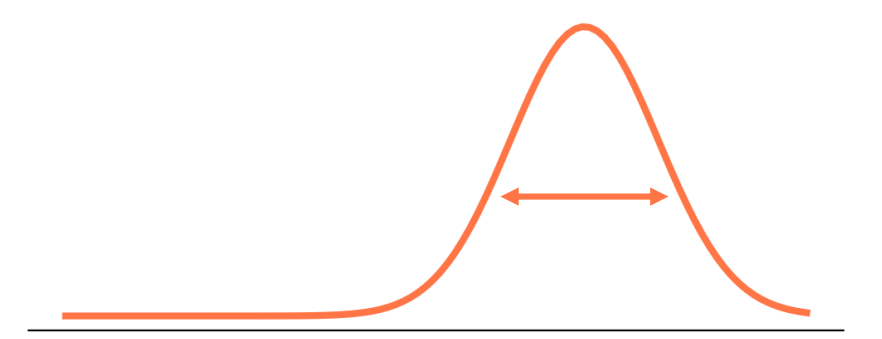
Projection

Marginal Distributions



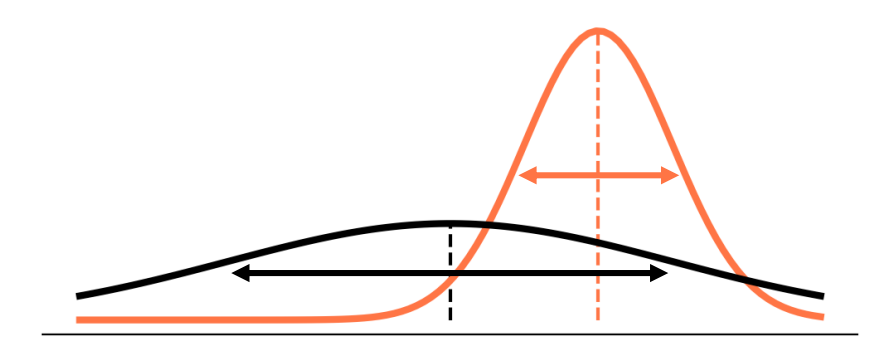
Inverse-Variance Importance

$$\frac{1}{Var}$$



KL Divergence Importance

$$D_{KL}$$



Importance Scores

- Inverse-Variance** measures spread of feature values.
 - KL Divergence** measures difference to data distribution.
- Approximated with Gaussians to distinguish features.

Procedure

- Trained random forests and computed explanations on test sets for 20 splits.
- Computed correlations to baselines over all features and test samples.
- Approximated gradient using average of 10 ReLU networks.

Baselines

Compared to absolute values of common explanations:

- Gradients** convey local sensitivity to input perturbations.
- SHAP values** determine the contribution of each feature to the function value.

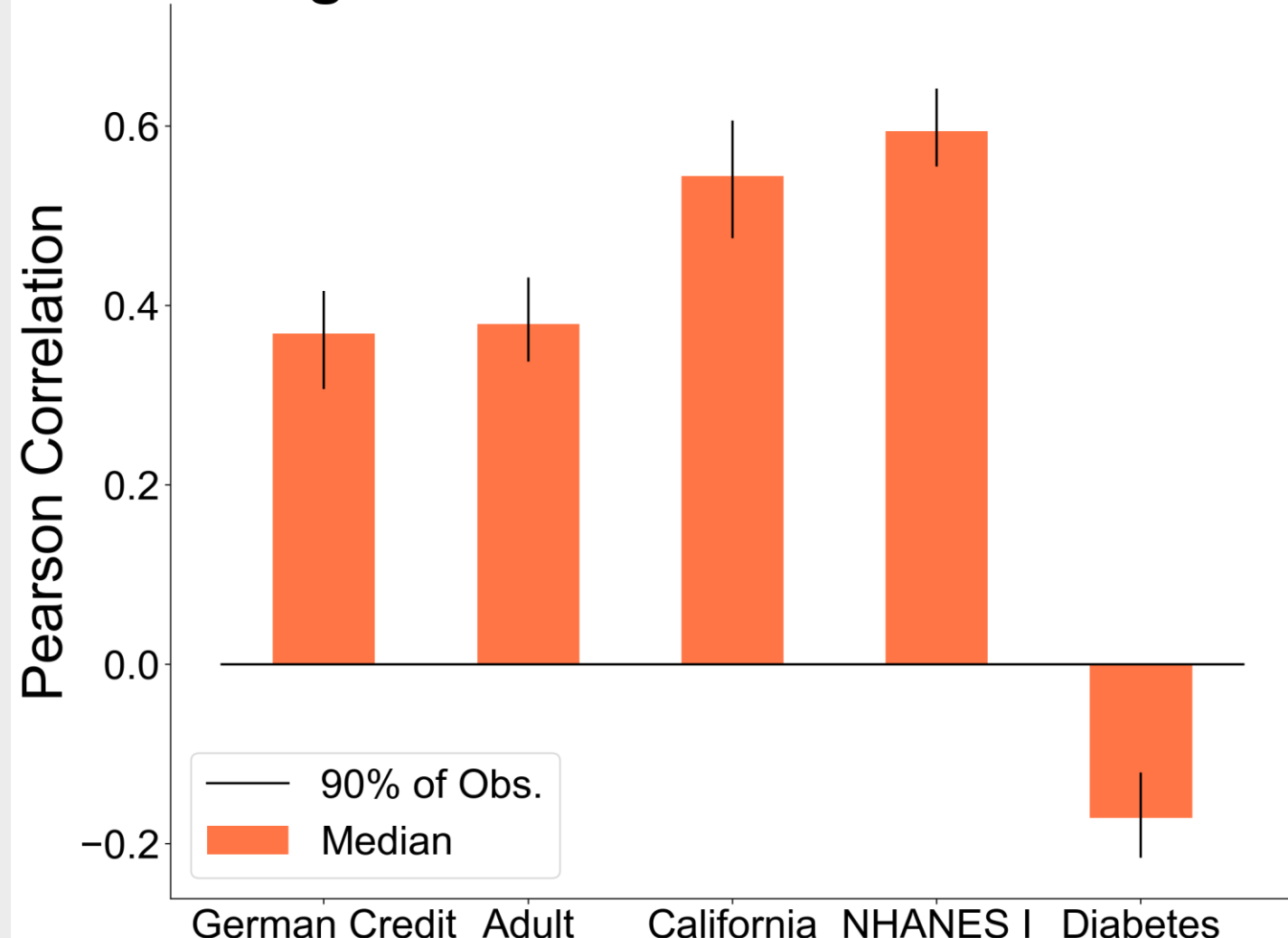
Conclusion

- Inconsistent relationship to baselines.
- Need theoretical results to justify usage.
- Potential for combining feature-based and instance-based explanations.

Acknowledgements

Kai Lüdemann is supported by the Konrad Zuse School of Excellence in Learning and Intelligent Systems (ELIZA) through the DAAD programme Konrad Zuse Schools of Excellence in Artificial Intelligence, sponsored by the Federal Ministry of Education and Research. We would also like to thank Dr. Michael Knaus and Henri Pfleiderer for consultation.

Log Inverse-Variance vs. Gradient



KL Divergence vs. SHAP Value

